

Predicting NBA Player Positions Using Supervised and Unsupervised Learning Techniques

A Data Science and Machine Learning Analysis of NBA Player Performance

Author: Waleed Farrakh

Techniques Used: K-Nearest Neighbors (KNN) • Support Vector Machines (SVM) • K-Means Clustering • Hierarchical Clustering

Tools: R • Tidyverse • caret • e1071 • cluster • factoextra

Dataset: NBA Player-Season Statistics (Stathead – Custom)

Research Question: Can NBA player statistical profiles accurately predict whether a player is a guard or non-guard, and do natural player groupings emerge from performance statistics alone?

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1. Introduction

The use of data analytics in professional basketball has grown significantly over the past decade, transforming how teams evaluate player performance, construct rosters, and make strategic decisions. Traditional box score statistics, while informative, often fail to fully capture a player's impact due to differences in pace, playing time, and team context. To address these limitations, advanced metrics such as per-75 possession statistics have become widely used. These metrics normalize player performance by accounting for pace and opportunity, allowing for more meaningful comparisons across players, seasons, and roles.

Per-75 possession statistics provide a standardized framework for analyzing key aspects of basketball performance, including scoring efficiency, shooting tendencies, rebounding, and playmaking. These features are closely tied to the functional roles that players occupy on the court. For example, guards are typically associated with higher assist rates, perimeter shooting, and ball-handling responsibilities, while non-guards (forwards and centers) tend to contribute more in rebounding, interior scoring, and defensive presence. However, modern basketball has evolved toward greater positional flexibility, with many players occupying hybrid roles (e.g., G-F or F-C), making it less clear whether traditional position labels can be accurately inferred from statistical profiles alone.

This project aims to investigate whether player statistical data can be used to reliably classify NBA players into positional categories. Specifically, the analysis focuses on distinguishing between guards and non-guards using supervised learning techniques. In addition, unsupervised learning methods are applied to explore whether natural groupings of players emerge based solely on their statistical characteristics, without the use of predefined labels.

The dataset used in this study consists of 600 NBA player-season observations and 32 variables. These variables include box score statistics (such as points, rebounds, and assists), shooting efficiency measures (such as field goal percentage, true shooting percentage, and effective field goal percentage), and other advanced performance metrics. Prior to analysis, the dataset was cleaned by removing irrelevant columns and addressing missing values. In particular, missing values in three-point percentage (3P%) were replaced with 0, reflecting players who did not attempt any three-point shots, ensuring that all observations could be retained for modeling.

The central research question guiding this analysis is:

Can NBA player statistical profiles accurately predict whether a player is a guard or non-guard?

To complement this, clustering methods are also used to examine whether players naturally group into distinct categories that align with positional roles. Together, these approaches provide insight into both the predictive power of statistical features and the underlying structure of player performance in modern basketball.

2. Methodology

This analysis uses both supervised and unsupervised learning methods to answer the research question. Supervised learning is used to predict player position, while unsupervised learning is used to identify natural groupings in the data.

2.1 K-Nearest Neighbors

KNN is a non-parametric, distance-based classification method. For a given observation, the algorithm identifies the K closest training points based on a distance metric (typically Euclidean distance) and assigns the class that appears most frequently among those neighbors.

Prediction rule:

$$\hat{y}(x) = \text{majority class among the } K \text{ nearest neighbors of } x$$

Distance is computed as:

$$\text{distance}(x_i, x_j) = \sqrt{\sum_p (x_{ip} - x_{jp})^2}$$

KNN was chosen because it provides a simple baseline model and does not assume any specific functional form between predictors and the response.

Advantages:

- Simple and intuitive
- No assumptions about data distribution

Disadvantages:

- Sensitive to scaling
- Performance depends on choice of K
- Computationally expensive for large datasets

2.2 Support Vector Machine

SVM is a classification method that finds an optimal hyperplane that separates classes by maximizing the margin between them.

The decision boundary is defined as:

$$w^T x + b = 0$$

For non-linear data, SVM uses a kernel function to map data into a higher-dimensional space. In this analysis, the radial basis function (RBF) kernel is used:

$$K(x_i, x_j) = \exp(-\gamma * \|x_i - x_j\|^2)$$

The final decision function is:

$$f(x) = \text{sign}(\text{sum over } i \text{ of } \alpha_i * y_i * K(x_i, x) + b)$$

SVM was chosen because it can model complex, non-linear relationships and often performs well in high-dimensional settings.

Advantages:

- Effective in high-dimensional spaces
- Can model complex decision boundaries

Disadvantages:

- Requires tuning of cost and gamma parameters
- Less interpretable than simpler models

2.3 K-Means Clustering

K-Means clustering partitions observations into K clusters by minimizing the within-cluster sum of squares.

Objective function:

$$\text{Minimize sum over } k \text{ of sum over } i \text{ in cluster } k \text{ of } \|x_i - \mu_k\|^2$$

where μ_k is the centroid of cluster k.

K-Means was chosen because it provides a simple and efficient way to identify group structure in the data.

Advantages:

- Simple and fast
- Easy to interpret

Disadvantages:

- Requires pre-specifying K
- Sensitive to initialization
- Assumes spherical clusters

2.4 Hierarchical Clustering

Hierarchical clustering builds a tree structure (dendrogram) by iteratively merging clusters based on a linkage criterion.

In this analysis, average linkage is used, where the distance between two clusters A and B is defined as:

$$d(A, B) = \text{average of all pairwise distances between points in A and points in B}$$

Clusters are merged step-by-step based on the smallest inter-cluster distance until all observations belong to a single cluster.

Hierarchical clustering was chosen because it does not require pre-specifying the number of clusters and provides a full hierarchy of group relationships.

Advantages:

- Does not require pre-specifying K
- Provides a full cluster hierarchy

Disadvantages:

- Computationally expensive
- Sensitive to linkage method

3. Data Analysis

3.1 Data Preprocessing and Thought Process

Several preprocessing steps were performed before modeling:

- Removed non-numeric variables such as Player, Team, and Season, as these are identifiers or categorical variables that are not directly meaningful for distance-based methods
- Retained only numerical features to ensure compatibility with KNN, SVM, and clustering algorithms
- Created a binary response variable (Guard vs NonGuard) based on position labels
- Replaced missing values in 3P% with 0 to reflect players with no three-point attempts
- Standardized all predictors using centering and scaling

Scaling was necessary because KNN, SVM, and clustering methods rely on distance calculations, and variables on different scales would otherwise dominate the results.

Potential multicollinearity was considered, particularly between TS% and eFG%, which measure similar aspects of shooting efficiency. While these variables are correlated, both were retained to preserve interpretability and capture different aspects of scoring efficiency.

No additional feature selection was performed in order to retain the full statistical profile of each player, as all variables were considered relevant from a basketball perspective.

3.2 Supervised Learning Results

The dataset was randomly split into 80% training and 20% testing sets using `set.seed(123)` to ensure reproducibility.

KNN Results

- Tested K values from 1 to 25
- Accuracy was evaluated on the test set for each K
- Optimal K selected: K = 5 (best balance of performance and stability)
- Test Accuracy: 89.08%

The relationship between K and test accuracy is shown in Figure 1, which was used to guide the selection of the optimal value of K.

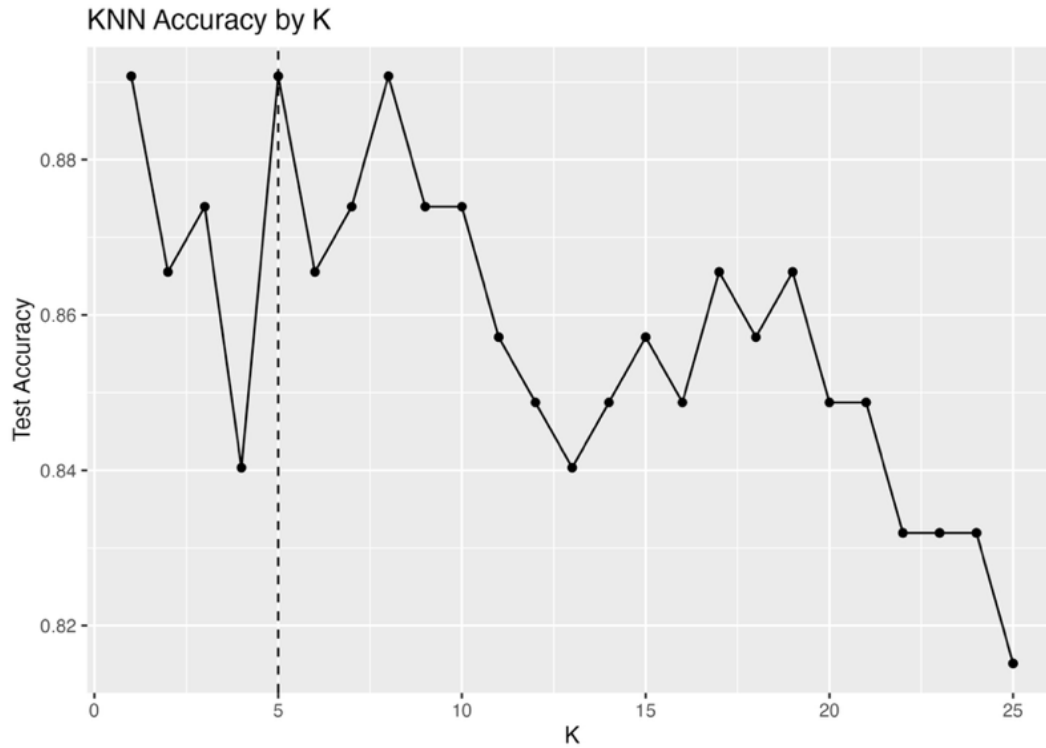


Figure 1: KNN accuracy as a function of K

SVM Results

- Tuned parameters using cross-validation:
 - cost = 10
 - gamma = 0.1
- These values produced the highest validation performance
- Test Accuracy: 89.92%

Model Comparison

Model	Test Accuracy
KNN	89.08%
SVM	89.92%

The SVM model slightly outperformed KNN, indicating a stronger ability to capture non-linear relationships between player statistics and positional roles.

3.3 Unsupervised Learning Results

K-Means Clustering

- Tested K values from 2 to 10
- Optimal K selected: K = 2 based on maximum silhouette score
- Silhouette Score: 0.174

Hierarchical Clustering

- Tested linkage methods: complete, average, and ward
- Best configuration:
 - Linkage: average
 - K = 2
- Silhouette Score: 0.208

Clustering Comparison

Model	Silhouette
K-Means	0.174
Hierarchical	0.208

Hierarchical clustering performed slightly better, although both methods produced relatively low silhouette scores.

To evaluate clustering quality, external class labels (GuardClass) were compared to cluster assignments. While some separation was observed, both clusters contained a mix of guards and non-guards, indicating imperfect alignment.

3.4 Interpretation of Clustering

Cluster analysis showed partial separation between guards and non-guards:

- One cluster was dominated by guards
- The other contained more non-guards
- However, significant overlap remained

This suggests that player roles are not strictly separable based on statistical features alone, likely due to hybrid positions and overlapping responsibilities in modern basketball. The clustering structure is visualized in **Figure 2**, which shows the K-means cluster assignments in two dimensions.

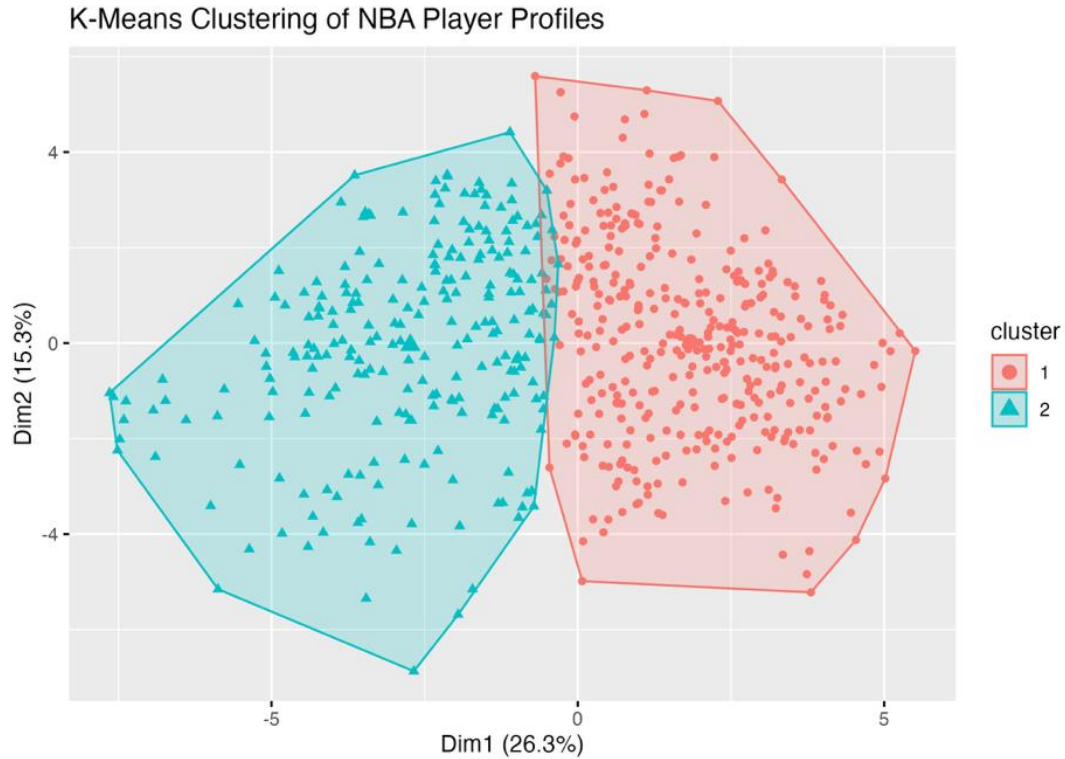


Figure 2: K-means clustering visualization

3.5 Final Model on Full Dataset

The best-performing model (SVM) was fitted on the full dataset.

Results:

- Accuracy: 100%
- Support Vectors: 382 (209 Guard, 173 NonGuard)

Figure 3 illustrates the SVM decision boundary using selected predictors, showing how the model separates guard and non-guard players in feature space.

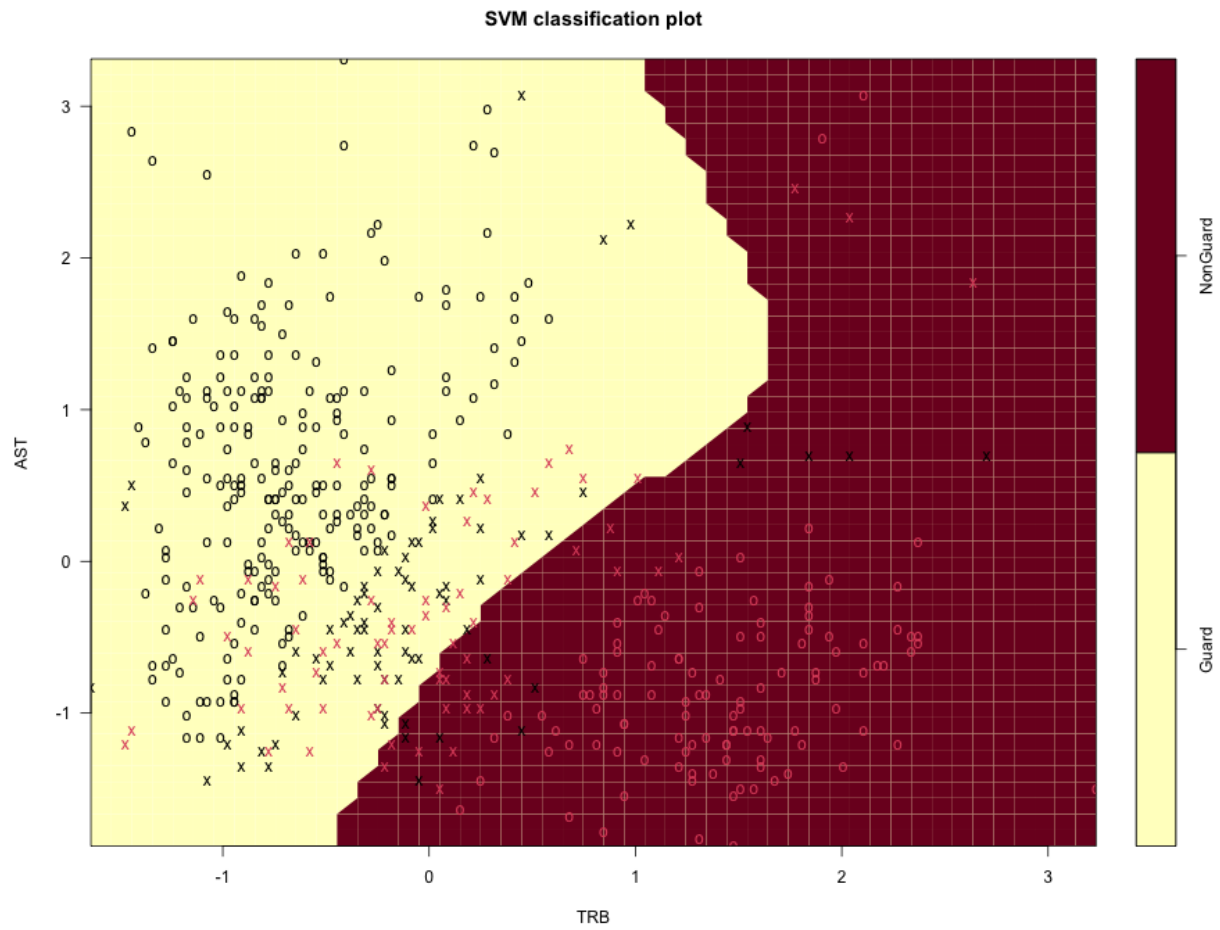


Figure 3: SVM decision boundary plot

For clustering visualization, a hierarchical clustering dendrogram was also generated.

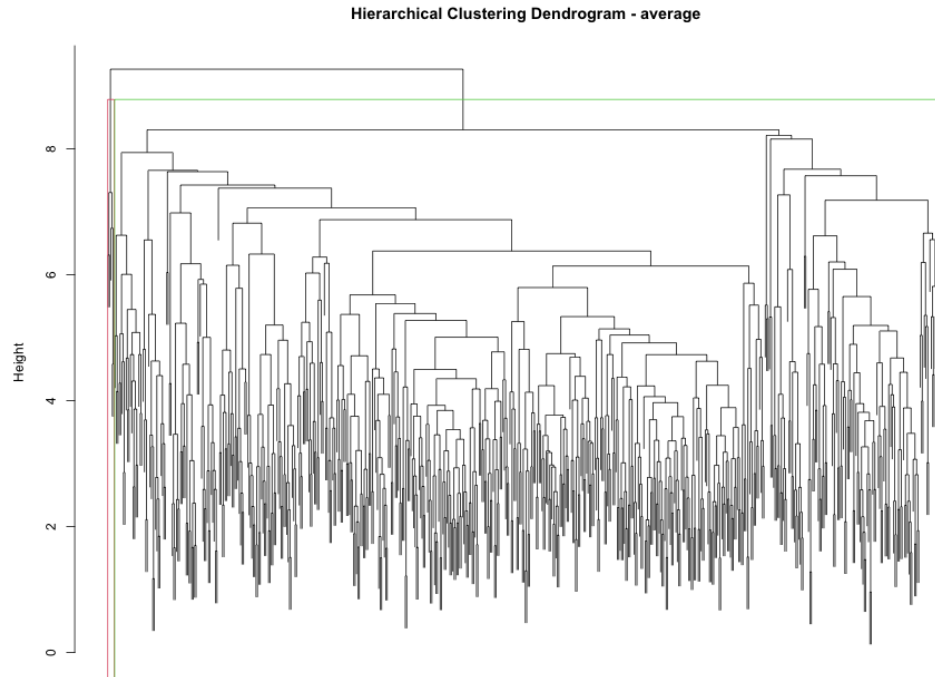


Figure 4: Hierarchical clustering dendrogram

These visualizations provide insight into how observations are grouped and separated.

This perfect classification accuracy reflects overfitting, since the model is evaluated on the same data used for training.

3.6 Overall Interpretation

The results indicate that NBA player statistical profiles can be used to predict positional roles with reasonably high accuracy (~90%), suggesting that measurable performance metrics capture key differences between guards and non-guards.

However, clustering results show weak separation, with low silhouette scores and overlapping clusters. This indicates that player roles are not naturally divided into clearly distinct groups, reflecting the increasing positional flexibility in modern basketball.

Overall, while supervised learning achieves strong predictive performance, unsupervised methods reveal that the underlying structure of player roles is more continuous than discrete.

4. Conclusion

This study demonstrates that NBA player statistical profiles can be used to predict positional roles with a relatively high degree of accuracy. The SVM model achieved the best performance, with a test accuracy of approximately 89.92%, outperforming the KNN model. This indicates that player statistics such as shooting efficiency, assists, and rebounds capture meaningful differences between guards and non-guards.

In contrast, clustering methods produced relatively low silhouette scores, indicating weak natural separation between groups. Although some structure was observed, the clusters did not perfectly align with positional labels. This suggests that modern basketball positions are fluid and overlapping, and that players cannot always be clearly divided into distinct categories based solely on statistical features.

Several difficulties were encountered during the data analysis process. One challenge was handling missing values, particularly for three-point percentage, which required careful consideration to avoid removing valuable observations. Another difficulty was the presence of correlated variables, such as TS% and eFG%, which may introduce redundancy into the model. Additionally, the presence of hybrid player positions (e.g., G-F) made classification less clear and contributed to overlap in clustering results. Finally, the relatively low silhouette scores indicate that the underlying structure of the data is not strongly separated, making clustering inherently challenging.

To improve this analysis, several approaches could be considered. More advanced models, such as ensemble methods or neural networks, could potentially improve predictive performance. Incorporating additional contextual variables, such as player role, team system, or defensive metrics, may also provide a more complete representation of player performance. Furthermore, applying feature selection or dimensionality reduction techniques could help reduce redundancy and improve model interpretability. For clustering, alternative methods such as density-based clustering may better capture the structure of the data.

Overall, while supervised learning models were effective in predicting player positions, unsupervised methods revealed that the distinction between roles is not sharply defined. This reflects the evolving nature of basketball, where positional roles are increasingly flexible and overlapping.

5. References

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